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Using machine learning to identify individuals at elevated risk of diabetes from low-cost complete blood count data for opportunistic screening

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Type 2 diabetes and prediabetes remain substantially underrecognized, highlighting the need for practical screening approaches based on routinely available data. This study aimed to develop and temporally validate a low-cost complete blood count (CBC)-based machine learning model for identifying individuals at elevated risk of diabetes in a routine health examination setting. We conducted a retrospective single-center study using electronic health record data from Shenzhen University General Hospital. A development cohort of 70,725 individuals examined between 2018 and 2023 was used for model development with stratified 5-fold cross-validation, and an independent temporal validation cohort of 26,650 individuals examined in 2024 was used for final evaluation. The outcome was defined as a binary diabetes-risk label based on HbA1c, with HbA1c < 5.7% classified as non-risk and HbA1c ≥ 5.7% classified as diabetes-risk, including both prediabetes and HbA1c-defined diabetes. Candidate models included logistic regression, LightGBM, multilayer perceptron, Tabular ResNet, and an attention-based tabular neural network. In the development cohort, LightGBM achieved the best overall performance, with a mean AUROC of 0.821 ± 0.006 and a mean AUPRC of 0.628 ± 0.010. In the independent 2024 temporal validation cohort, LightGBM again performed best, with an AUROC of 0.791 and an AUPRC of 0.699. Calibration analysis showed preserved risk ranking but some overestimation of absolute risk, and subgroup analysis revealed performance heterogeneity across sex, age, and BMI strata. These findings suggest that CBC-based machine learning, particularly LightGBM, may provide a practical approach for opportunistic identification of individuals at elevated risk of diabetes and support further confirmatory testing in routine health examination settings.

Type 2 diabetes is one of the most common metabolic disorders and is characterized by chronic hyperglycemia resulting from impaired insulin secretion by pancreatic β-cells and insulin resistance in peripheral tissues[1,2]. Persistent hyperglycemia can lead to multisystem damage and contributes to increased susceptibility to infection, cancer progression, and other pathophysiological consequences through mechanisms including oxidative stress, activation of the polyol and protein kinase C pathways, and formation of advanced glycation end products[3]. In 2024, diabetes caused 3.4 million deaths worldwide, with an additional 530,000 kidney disease deaths attributable to diabetes; high blood glucose also accounted for approximately 11% of cardiovascular deaths[4]. According to the American Diabetes Association, the estimated total economic burden of diagnosed diabetes in the United States reached $412.9 billion in 2022[5]. Early identification and intervention may delay disease progression, reduce the risk of complications, improve quality of life, and lessen the healthcare burden[5].

The global burden of diabetes continues to rise. According to the 11th edition of the IDF Diabetes Atlas, 589 million adults aged 20–79 years were living with diabetes worldwide in 2024, equivalent to 1 in 9 adults, and this number is projected to increase to 783 million by 2045. Despite this, an estimated 252 million people worldwide, or 43% of all cases, remain undiagnosed[4]. China bears the largest absolute burden, with approximately 140 million people living with diabetes and about 72.8 million cases remaining undiagnosed[6]. National data further indicate that the awareness rate of diabetes in China is only 36.5%[7]. Due to limited access to healthcare

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services and the constrained capacity of existing health systems[8], more practical and scalable approaches are needed to improve the identification of individuals with unrecognized diabetes or elevated diabetes risk.

In addition to formal population-based screening, opportunistic screening may offer an important complementary strategy. Opportunistic screening refers to the identification of diseases or health risks during routine contacts with the healthcare system, such as health examinations, outpatient visits, or other clinical encounters, without a predefined invitation-based screening program. This strategy is commonly applied to conditions like breast cancer[9] and osteoporosis[10]. For diabetes, opportunistic screening may be particularly valuable in settings where routine laboratory data are already being collected but glycemia-specific tests are not universally prioritized. In such contexts, prediction models based on routinely available non-glucose indicators may help identify individuals who warrant further confirmatory testing. Machine learning has been extensively applied to diabetes-related predictions[11]. Methodologically, previous studies have progressed from conventional approaches such as logistic regression, decision trees, and support vector machines to more complex ensemble strategies. In terms of predictors, many studies have incorporated direct glycemic indices or broader clinical and lifestyle information, such as dietary behavior, exercise habits[12], family history, salt preference, physical activity, and work stress[13]. However, such information may not always be consistently available in routine clinical workflows. A model based on low-cost and commonly available laboratory indicators may therefore have practical advantages for opportunistic screening.

Complete blood count (CBC) testing is one of the most commonly performed laboratory investigations in clinical practice[14]. Prior evidence suggests that diabetes may alter blood cell structure and function through hyperglycemia, oxidative stress, and inflammatory mechanisms, thereby affecting hematologic parameters[15]. Previous studies have explored the value of CBC-related markers for diabetes prediction. One study incorporated C-reactive protein as a major inflammatory indicator, which is less routinely available than CBC[16], whereas another study relied on relatively simple models, a smaller sample size, and limited validation[17]. Although these studies support the feasibility of using CBC-derived information for diabetes-related risk assessment, further work is needed using larger real-world datasets, more rigorous validation strategies, and clinically practical modeling frameworks.

Accordingly, the present study used electronic health record data from Shenzhen University General Hospital to develop and temporally validate a machine learning model based on CBC features and basic demographic variables for identifying individuals at elevated risk of diabetes in a routine health examination setting. Because the intended use of the model was opportunistic screening rather than definitive diagnosis, the study focused on identifying individuals who may warrant further confirmatory testing. The diabetes-risk group in this study included both prediabetes and HbA1c-defined diabetes. We further compared representative model families, including logistic regression, tree-based boosting, and neural-network-based tabular models, and evaluated their performance using a temporally independent 2024 cohort. By doing so, we aimed to assess the feasibility of a low-cost CBC-based screening approach within a real-world single-center setting.

Methods

Study design

This study was designed as a retrospective observational study to develop and validate a low-cost complete blood count (CBC)-based machine learning model for identifying individuals at elevated risk of diabetes. The overall workflow is illustrated in Fig. 1. To provide a clinically meaningful assessment of model generalizability, we adopted a temporal validation design rather than a random split across all years. Specifically, data collected between 2018 and 2023 were used to construct the development cohort, whereas data collected in 2024 were reserved as an independent temporal validation cohort. The development cohort was used for model development, internal validation, and model selection, while the temporal validation cohort was used exclusively for final performance evaluation.

This study aimed to identify both individuals with established diabetes and those with potentially abnormal glycemic status who may benefit from further diabetes screening. The diabetes-risk group in this study included both prediabetes and HbA1c-defined diabetes, because the intended use was opportunistic screening for individuals who may warrant further T2D evaluation. Because the intended use of the model was screening rather than definitive diagnosis, emphasis was placed on identifying individuals warranting further confirmatory testing.

Data source and participants

Data were obtained from routine health examination records. Individuals were eligible for inclusion if both CBC measurements and glycated hemoglobin (HbA1c) results were available from the same examination encounter. To ensure comparability between the development and temporal validation datasets, only variables available in both datasets were considered for model construction.

Participants missing variables required for outcome labeling were excluded. For predictor variables, implausible values were handled during data cleaning and retained as missing when appropriate for subsequent imputation. In addition, implausible laboratory values identified according to prespecified clinical rules were handled during the cleaning procedure described below. After applying the inclusion and exclusion criteria, 70,725 individuals were included in the development cohort and 26,650 individuals were included in the temporal validation cohort.

Outcome definition

The study outcome was defined as a binary diabetes-risk label based on HbA1c levels. According to the study design, individuals with HbA1c < 5.7% were classified as the non-risk group, whereas those with HbA1c ≥ 5.7% were classified as the diabetes-risk group. The latter included both individuals with prediabetes (HbA1c 5.7%–

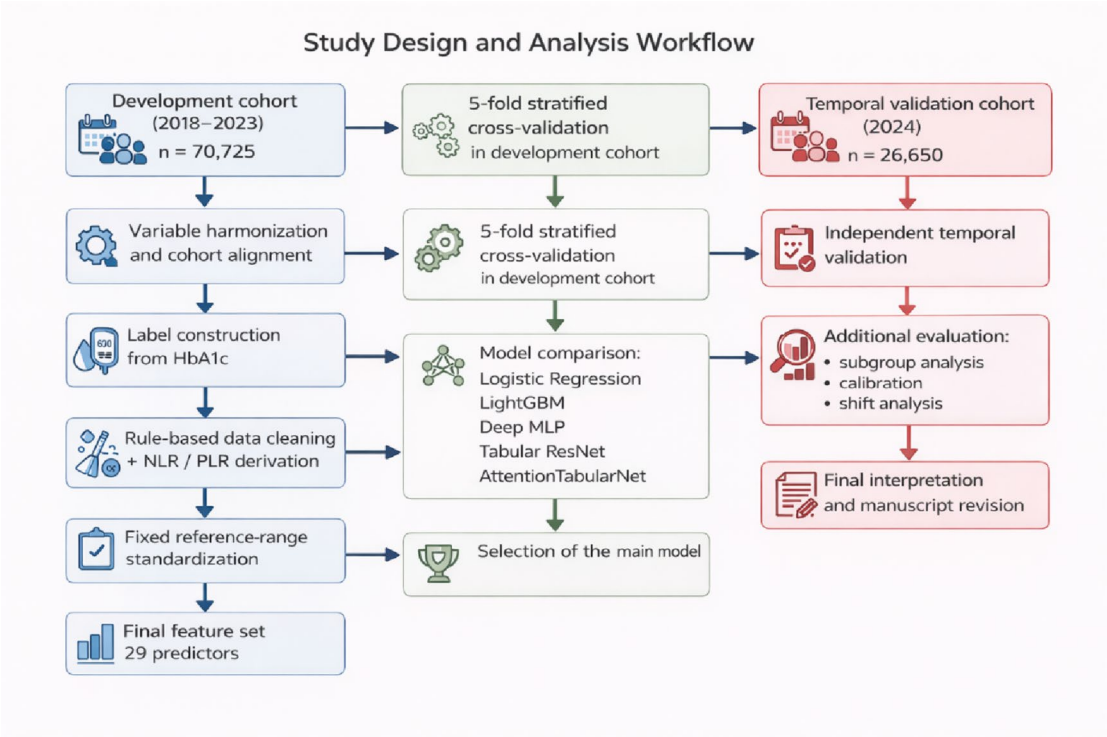


Fig. 1. Study design flowchart.

6.4%) and those meeting the HbA1c criterion for diabetes (HbA1c ≥ 6.5%). In the development cohort, 18,443 individuals were classified into the diabetes-risk group and 52,282 into the non-risk group.

This outcome definition was selected to match the intended clinical use of the model as a low-cost opportunistic screening tool based on CBC data. Rather than aiming to establish a definitive diagnosis, the model was designed to identify individuals who may already have diabetes or who may be at sufficiently elevated risk to warrant further confirmatory testing. The same outcome definition was applied consistently in both the development and temporal validation cohorts. To avoid target leakage, HbA1c was used only for outcome labeling and was excluded from the predictor set.

Variable harmonization

Because the datasets were collected over different calendar periods, variable names, formats, and measurement expressions were harmonized before analysis. Equivalent laboratory variables recorded under different naming conventions across years were manually mapped into a unified feature set, and relevant variables were converted into consistent numeric formats where applicable.

Sex coding was standardized across datasets, and only predictors that could be consistently aligned between the 2018–2023 and 2024 cohorts were retained for downstream analysis. This step ensured that the same predictor definitions were used during both model development and temporal validation.

Data cleaning

A rule-based data cleaning strategy was used to identify implausible or erroneous values. These rules were defined a priori according to clinical knowledge and laboratory reference standards. Variables screened in this procedure included demographic characteristics and CBC-related laboratory parameters. Rather than removing all records containing implausible values, values falling outside the prespecified plausible range were set to missing whenever appropriate, allowing the corresponding individual to be retained for subsequent processing.

This approach reduced the influence of obvious measurement or entry errors while preserving as much information as possible from the original dataset. The same prespecified cleaning rules were applied consistently to both the development and temporal validation cohorts.

Feature engineering

To enrich the predictor set with clinically relevant inflammatory indicators, two derived features were calculated from CBC parameters: the neutrophil-to-lymphocyte ratio (NLR) and the platelet-to-lymphocyte ratio (PLR). These markers have been widely used to reflect systemic inflammatory status and may be associated with metabolic dysregulation.

The derived variables were calculated after data cleaning to reduce the propagation of implausible raw values. If the denominator required for ratio calculation was missing or non-positive, the derived feature was treated as missing.

Reference-range-based standardization

To improve model robustness across cohorts and enhance clinical interpretability, laboratory variables were standardized using fixed clinical reference ranges rather than sample-dependent normalization methods. For each laboratory parameter with an established reference interval, the observed value was transformed relative to the lower and upper bounds of the corresponding normal range.

This strategy has two advantages. First, it is clinically interpretable because the transformed value reflects the extent to which the original measurement deviates from the normal reference interval. Second, because the transformation is based on prespecified reference standards rather than cohort-specific sample statistics, it is less sensitive to temporal distribution shifts and does not introduce information leakage from validation or test data.

Variables such as age, BMI, NLR, and PLR were retained in their original scale or handled according to the model-specific preprocessing pipeline.

Predictor set

The final predictor set consisted of demographic variables, derived inflammatory indicators, and CBC laboratory variables after harmonization and preprocessing. Age, BMI, sex, NLR, and PLR were included as key predictors. CBC parameters were represented in their reference-range-standardized form where applicable. Only variables available in both time periods and processed in a consistent manner were entered into model development.

Model development

Model development was conducted using the 2018–2023 development cohort. Multiple machine learning algorithms for structured tabular data were evaluated, including logistic regression, Light Gradient Boosting Machine (LightGBM), multilayer perceptron (MLP), Tabular ResNet, and an attention-based tabular neural network. Logistic regression was included as an interpretable baseline model, while the remaining methods represented tree-based and deep learning approaches for tabular prediction.

All candidate models were evaluated under a unified framework to enable fair comparison. Model selection was based on validation performance in the development cohort, after which the best-performing model was retrained using the full development cohort and evaluated in the independent temporal validation cohort.

Cross-validation strategy and missing data handling

Internal model development was performed using stratified 5-fold cross-validation within the development cohort. The dataset was partitioned into five folds while preserving the proportion of individuals in the diabetes-risk and non-risk groups. In each iteration, four folds were used for training and one fold was used for validation.

Missing values were handled using median imputation. During cross-validation, the imputation procedure was fitted using only the training folds and then applied to the corresponding validation fold to avoid data leakage. After model selection, the same preprocessing strategy was refit on the full development cohort before application to the temporal validation cohort.

Leakage control

Particular attention was paid to preventing data leakage throughout the modeling workflow. All preprocessing steps that required parameter estimation from data, including median imputation and any model-specific scaling procedures, were fitted using only the training portion of each fold and then applied to the corresponding validation fold. The independent 2024 temporal validation cohort was completely isolated from all stages of model training, hyperparameter tuning, and model selection.

In contrast, preprocessing steps based on prespecified external knowledge rather than data fitting, such as variable harmonization rules, rule-based cleaning criteria, and fixed reference-range transformation, were defined independently of validation and test outcomes and applied consistently across cohorts.

Temporal validation

After model selection in the development phase, the final selected model was retrained using the full 2018–2023 development cohort and then evaluated on the independent 2024 temporal validation cohort. Because this cohort was not used in any step of model development, it provided an unbiased estimate of model performance in a temporally distinct population.

Temporal validation was especially important in this study because the intended application was opportunistic screening in future health examination populations, where shifts in participant composition and laboratory patterns may occur over time.

Evaluation metrics

Model performance was summarized using the area under the receiver operating characteristic curve (AUROC), the area under the precision-recall curve (AUPRC), accuracy, F1 score, sensitivity, specificity, and Brier score. AUROC was used to evaluate overall discrimination across thresholds, whereas AUPRC was additionally reported because it is informative under class imbalance. Accuracy, F1 score, sensitivity, and specificity were used to characterize threshold-dependent classification performance, and the Brier score was used to assess the overall accuracy of probabilistic predictions.

Additional analyses

To further assess model performance and robustness, additional analyses were conducted, including subgroup analysis, calibration analysis, and distribution shift analysis. Subgroup analyses were used to examine whether

model performance remained stable across demographic strata. Calibration analysis was performed to evaluate the agreement between predicted probabilities and observed outcome rates. Distribution shift analysis was conducted using standardized mean differences (SMDs) to quantify temporal differences in variable distributions between the development and temporal validation cohorts.

Statistical analysis

Descriptive statistics were used to summarize baseline characteristics. Continuous variables were presented as mean ± standard deviation, and categorical variables were presented as counts and percentages.

Statistical comparisons between candidate models in the development phase were based on fold-wise paired performance estimates obtained from the stratified 5-fold cross-validation procedure. Because each model was evaluated on the same validation folds, paired comparisons were conducted using the Wilcoxon signed-rank test. Performance in the independent 2024 temporal validation cohort was summarized descriptively.

All statistical tests were two-sided, and a p value < 0.05 was considered statistically significant.

Software and reproducibility

All analyses were conducted using Python and relevant data science and machine learning libraries. To improve reproducibility, the preprocessing workflow, model development procedures, and evaluation pipeline were implemented within a fixed analytical framework with controlled random seeds where applicable.

Results

Cohort characteristics

The final analytic dataset included 70,725 individuals in the 2018–2023 development cohort and 26,650 individuals in the 2024 temporal validation cohort (Fig. 1; Table 1). The proportion of individuals classified into the diabetes-risk group increased from 26.1% in the development cohort to 38.1% in the temporal validation cohort, indicating a substantially higher prevalence of elevated diabetes risk in the later cohort.

Variable	Development (2018–2023)	Temporal validation (2024)	SMD
Age	37.641 ± 10.589	39.073 ± 10.381	0.137
BMI	23.438 ± 3.475	24.018 ± 3.512	0.166
Sex	0.585 ± 0.493	0.643 ± 0.479	0.119
T2D Risk Category	0.261 ± 0.439	0.381 ± 0.486	0.259
NLR	1.750 ± 0.717	1.727 ± 0.705	-0.031
PLR	129.833 ± 41.017	128.218 ± 40.473	-0.040
WBC_stdref	0.456 ± 0.253	0.457 ± 0.286	0.003
RBC_stdref	0.582 ± 0.283	0.581 ± 0.270	-0.004
Hemoglobin_stdref	0.518 ± 0.274	0.535 ± 0.259	0.063
Hematocrit_stdref	0.604 ± 0.290	0.594 ± 0.271	-0.033
Neutrophil%_stdref	0.444 ± 0.224	0.432 ± 0.225	-0.051
Lymphocyte%_stdref	0.480 ± 0.242	0.488 ± 0.244	0.031
Monocyte%_stdref	0.563 ± 0.236	0.579 ± 0.235	0.069
Eosinophil%_stdref	0.276 ± 0.250	0.288 ± 0.261	0.050
Basophil%_stdref	0.631 ± 0.303	0.594 ± 0.285	-0.127
Neutrophil#_stdref	0.378 ± 0.255	0.373 ± 0.255	-0.021
Lymphocyte#_stdref	0.481 ± 0.271	0.488 ± 0.274	0.024
Monocyte#_stdref	0.659 ± 0.287	0.673 ± 0.280	0.049
Eosinophil#_stdref	0.273 ± 0.286	0.285 ± 0.276	0.042
Basophil#_stdref	0.653 ± 1.512	0.612 ± 0.494	-0.036
MCV_stdref	0.395 ± 0.346	0.379 ± 0.297	-0.051
MCH_stdref	0.365 ± 0.369	0.392 ± 0.319	0.078
MCHC_stdref	0.403 ± 0.318	0.491 ± 0.277	0.294
RDW-CV_stdref	0.518 ± 0.378	0.506 ± 0.318	-0.035
RDW-SD_stdref	0.287 ± 0.133	0.285 ± 0.130	-0.011
Platelet_stdref	0.595 ± 0.257	0.588 ± 0.252	-0.029
MPV_stdref	0.853 ± 0.235	0.772 ± 0.225	-0.349
PDW_stdref	0.488 ± 0.281	0.386 ± 0.280	-0.363
P-LCR_stdref	0.516 ± 0.246	0.429 ± 0.234	-0.361
PCT_stdref	0.840 ± 0.274	0.789 ± 0.262	-0.188

Table 1. Baseline characteristics of the development and temporal validation cohorts. Continuous variables are presented as mean ± standard deviation. SMD indicates the standardized mean difference between the temporal validation cohort and the development cohort.

Compared with the development cohort, participants in the 2024 temporal validation cohort were slightly older (39.073 ± 10.381 vs. 37.641 ± 10.589 years) and had a higher BMI (24.018 ± 3.512 vs. 23.438 ± 3.475 kg/m²) (Table 1). Standardized mean difference analysis further showed measurable temporal shifts in several variables, with the largest absolute shifts observed for PDW_stdref (SMD = -0.363), P-LCR_stdref (SMD = -0.361), MPV_stdref (SMD = -0.349), MCHC_stdref (SMD = 0.294), and the diabetes-risk label itself (SMD = 0.259) (Figure S4). These findings indicate that the 2024 cohort represented a temporally distinct population rather than a simple continuation of the earlier data distribution.

Model development in the 2018–2023 cohort

All five candidate models showed similar discrimination in stratified 5-fold cross-validation within the development cohort, with AUROC values ranging from 0.816 to 0.821 and AUPRC values ranging from 0.614 to 0.628 (Table 2). LightGBM achieved the best overall development performance, with a mean AUROC of 0.821 ± 0.006 and a mean AUPRC of 0.628 ± 0.010, whereas logistic regression yielded slightly lower values (AUROC 0.816 ± 0.005; AUPRC 0.614 ± 0.009). The corresponding out-of-fold ROC and precision-recall curves are shown in Figure S1 and Figure S2.

For threshold-dependent metrics, LightGBM achieved an accuracy of 0.746 ± 0.002, an F1 score of 0.602 ± 0.003, a recall of 0.738 ± 0.005, a specificity of 0.748 ± 0.002, and a Brier score of 0.171 ± 0.002 (Table 2). Deep MLP showed the highest specificity (0.923 ± 0.006) and lowest Brier score (0.141 ± 0.002), but this was accompanied by substantially reduced recall (0.441 ± 0.013), making it less suitable for a screening-oriented application. Overall, LightGBM provided the most balanced combination of discrimination and classification performance and was therefore selected as the main model.

Fold-wise paired comparisons against logistic regression showed a consistent numerical advantage for LightGBM in the development phase, but these differences did not reach conventional statistical significance under the Wilcoxon signed-rank test (AUROC, AUPRC, Brier score, and F1: $P = 0.0625$; Accuracy: $P = 0.1250$).

Independent temporal validation in 2024

In the independent 2024 temporal validation cohort, all models showed some decline in performance relative to the development cohort, although the ranking pattern remained broadly similar (Figs.2 and 3; Table 3). LightGBM again showed the best overall performance, with an AUROC of 0.791 and an AUPRC of 0.699. Tabular ResNet and Deep MLP performed similarly (AUROC 0.790 and 0.789; AUPRC 0.698 and 0.698, respectively), whereas logistic regression showed the lowest AUROC (0.786) and AUPRC (0.689).

For threshold-dependent performance in the temporal validation cohort, LightGBM achieved an accuracy of 0.720, an F1 score of 0.657, a recall of 0.705, a specificity of 0.729, and a Brier score of 0.185 (Table 3). Compared with logistic regression, LightGBM showed modest improvements in AUROC, AUPRC, F1 score, recall, and Brier score, while maintaining similar accuracy. By contrast, Deep MLP again displayed a high-specificity but low-recall profile, which was less aligned with the intended screening use of the model. Taken together, these results supported LightGBM as the main model for identifying individuals at elevated risk of diabetes in future cohorts.

Calibration and subgroup performance of the LightGBM model

Calibration analysis in the 2024 temporal validation cohort showed that the LightGBM model preserved appropriate risk ranking across deciles, but tended to overestimate absolute risk overall (Fig. 4). The mean predicted probability was 0.450, whereas the observed positive rate was 0.381. Across deciles of predicted risk, the observed positive rate increased from 0.058 in the lowest-risk decile to 0.832 in the highest-risk decile, indicating good monotonic separation despite imperfect calibration.

Subgroup analysis further showed that model performance was heterogeneous across sex, age, and BMI categories (Table4; Figure S3). AUROC was higher in females than in males (0.808 vs. 0.773), although AUPRC was higher in males, consistent with the higher positive rate in that subgroup. Performance also declined with age, with AUROC values of 0.754 in participants aged < 40 years, 0.730 in those aged 40–59 years, and 0.680 in those aged ≥ 60 years. By BMI category, AUROC was highest in participants with BMI < 24 kg/m² (0.786) and lowest in those with BMI ≥ 28 kg/m² (0.737), whereas AUPRC increased with BMI category. These findings suggest that although the model retained acceptable overall discrimination, its performance was influenced by subgroup-specific prevalence and case mix.

Model	AUROC	AUPRC	Accuracy	F1	Recall	Specificity	Brier Score
LightGBM	0.821 ± 0.006	0.628 ± 0.010	0.746 ± 0.002	0.602 ± 0.003	0.738 ± 0.005	0.748 ± 0.002	0.171 ± 0.002
Tabular ResNet	0.821 ± 0.006	0.626 ± 0.011	0.739 ± 0.010	0.601 ± 0.007	0.753 ± 0.012	0.734 ± 0.016	0.175 ± 0.005
Attention Tabular Net	0.821 ± 0.006	0.626 ± 0.010	0.739 ± 0.008	0.600 ± 0.005	0.751 ± 0.016	0.735 ± 0.016	0.175 ± 0.003
Deep MLP	0.820 ± 0.005	0.627 ± 0.010	0.797 ± 0.002	0.532 ± 0.008	0.441 ± 0.013	0.923 ± 0.006	0.141 ± 0.002
Logistic Regression	0.816 ± 0.005	0.614 ± 0.009	0.745 ± 0.001	0.599 ± 0.003	0.730 ± 0.007	0.750 ± 0.002	0.175 ± 0.002

Table 2. Performance of Five Models in the Development Cohort Using 5-Fold Cross-Validation. Data are presented as mean ± standard deviation across the 5 validation folds. Abbreviations: AUROC, area under the receiver operating characteristic curve; AUPRC, area under the precision-recall curve.

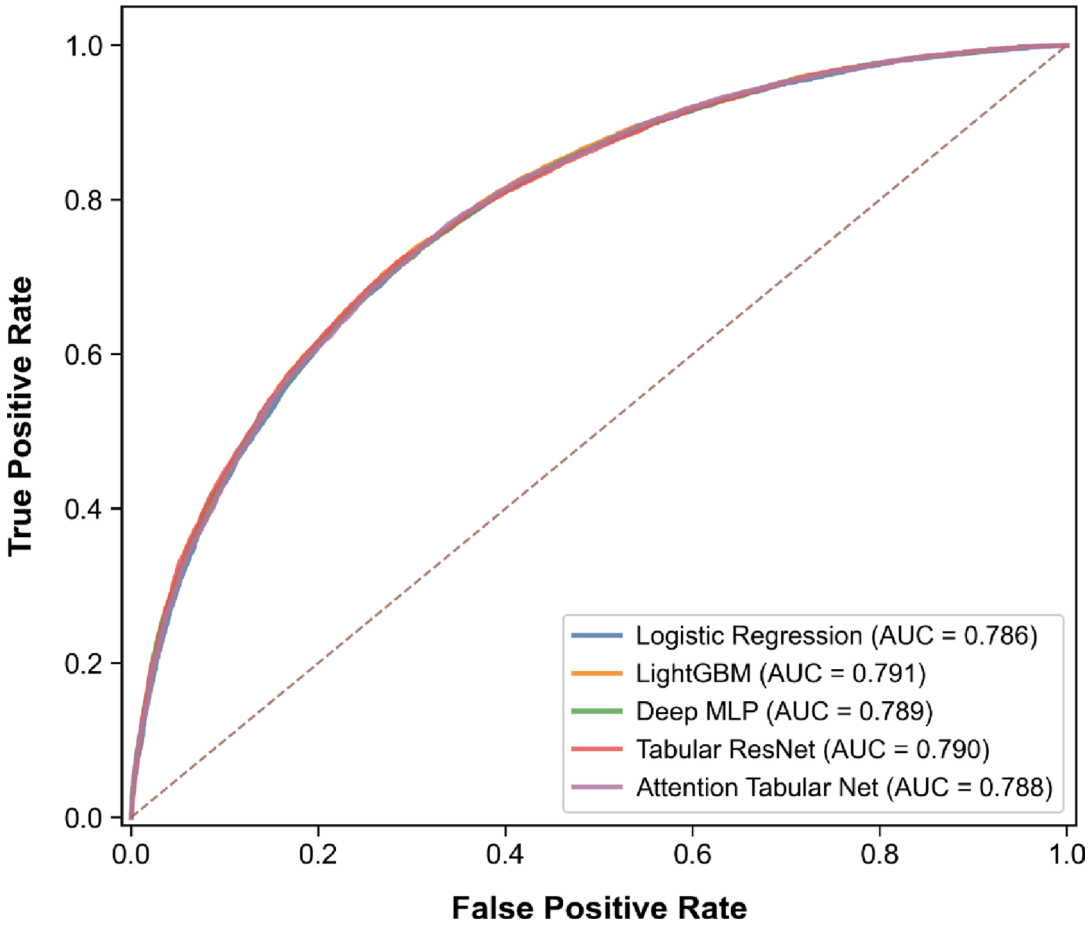


Fig. 2. ROC 2024 temporal validation refined.

Model	AUROC	AUPRC	Accuracy	F1	Recall	Specificity	Brier Score
LightGBM	0.791	0.699	0.720	0.657	0.705	0.729	0.185
Tabular ResNet	0.790	0.698	0.717	0.658	0.716	0.718	0.187
Deep MLP	0.789	0.698	0.720	0.514	0.389	0.924	0.191
Attention Tabular Net	0.788	0.694	0.715	0.654	0.708	0.720	0.186
Logistic Regression	0.786	0.689	0.718	0.649	0.685	0.739	0.187

Table 3. Performance of Five Models in the 2024 Temporal Validation Cohort. Abbreviations: AUROC, area under the receiver operating characteristic curve; AUPRC, area under the precision-recall curve.

Summary of the main findings

Overall, LightGBM demonstrated the most favorable and most balanced performance among the evaluated models in both the development and temporal validation cohorts. Although its numerical advantage over logistic regression was modest, it consistently outperformed the conventional baseline while maintaining a more screening-appropriate trade-off between sensitivity and specificity than Deep MLP. The model also retained acceptable discrimination in the independent 2024 cohort despite measurable temporal distribution shifts, although some degree of risk overestimation and subgroup heterogeneity remained, indicating that further calibration and population-specific refinement may be beneficial before real-world opportunistic screening deployment.

Discussion

This study developed and temporally validated a complete blood count (CBC)-based machine learning model for identifying individuals at elevated risk of diabetes in a routine health examination setting. Among the evaluated models, LightGBM showed the best overall balance between discrimination and screening-oriented classification performance. Although its numerical advantage over the other candidate models was modest,

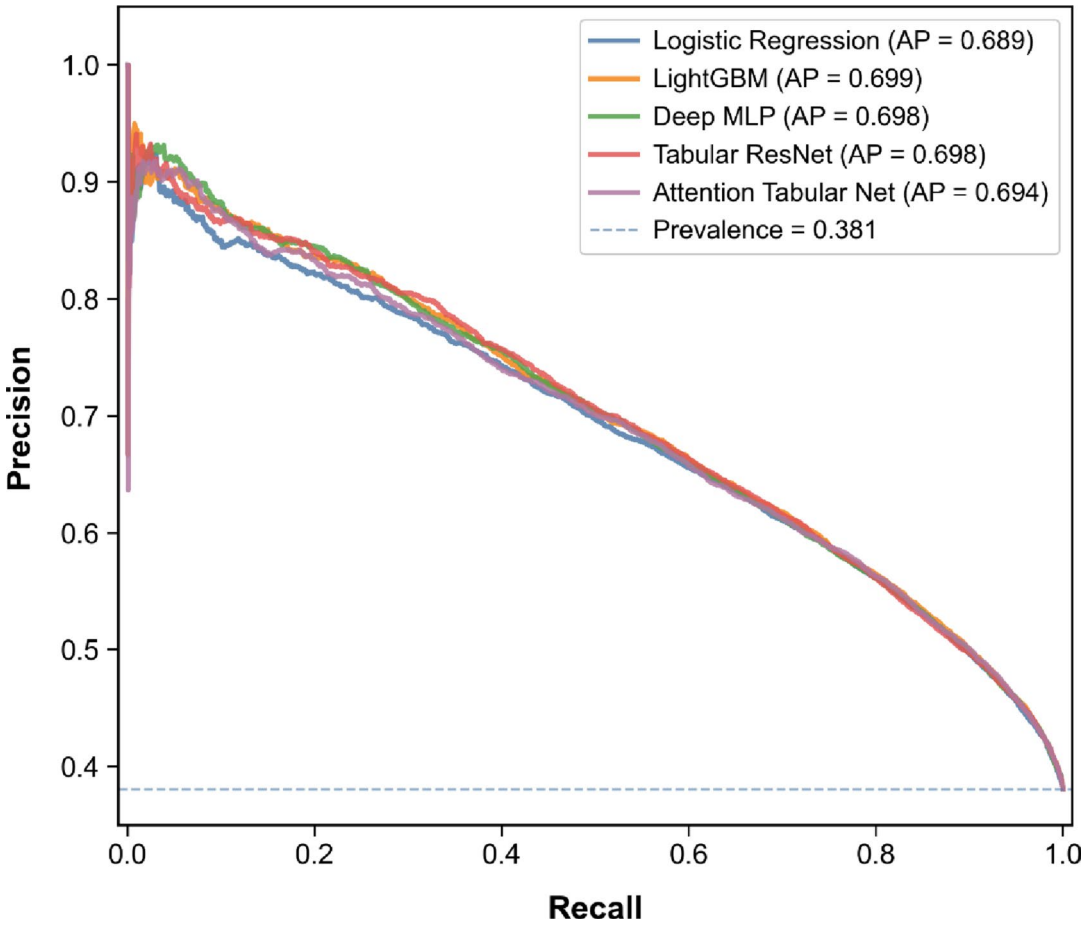


Fig. 3. PR 2024 temporal validation refined.

it remained the strongest-performing model in both the development and temporal validation cohorts and retained acceptable discrimination in the independent 2024 cohort.

An important feature of this study is that the prediction target was aligned with the intended use of the model. Rather than diagnosing overt diabetes alone, the model was designed to identify individuals at elevated diabetes risk, including both prediabetes and HbA1c-defined diabetes, who may benefit from further confirmatory testing. Given that CBC testing is inexpensive and routinely available in health examination settings, such a model may serve as a practical aid for opportunistic screening rather than as a stand-alone diagnostic tool.

LightGBM consistently outperformed logistic regression numerically and performed at least comparably to the evaluated neural-network-based tabular models. These findings suggest that for relatively structured CBC and demographic data, a tree-based boosting method may provide a more favorable balance between performance and complexity than either conventional linear modeling or more complex neural architectures. In contrast, Deep MLP showed a higher-specificity but lower-recall pattern, which is less desirable for a screening-oriented application.

Model performance declined modestly in the temporal validation cohort. However, this decline should be interpreted in the context of measurable temporal changes in both predictor distributions and outcome prevalence. The 2024 cohort was older, had a higher BMI, and included a higher proportion of individuals in the diabetes-risk group. Several hematologic variables, particularly platelet-related indices, also differed across periods. These findings suggest that temporal population shift likely contributed to the observed attenuation in performance.

Calibration and subgroup analyses provided additional insight. The LightGBM model preserved monotonic risk ranking across predicted-risk deciles, although it tended to overestimate absolute risk in the temporal validation cohort. In addition, model performance varied across sex, age, and BMI strata, indicating that subgroup-specific prevalence and case mix influenced apparent performance. These findings suggest that recalibration or subgroup-specific threshold adjustment may be useful before clinical deployment.

This study has several strengths, including a large real-world dataset, temporal validation, a leakage-controlled preprocessing framework, and comparison across representative model families. Several limitations should also be acknowledged. First, this was a retrospective single-center study, and broader generalizability remains uncertain. Second, the outcome was defined as elevated diabetes risk rather than overt diabetes alone, which is appropriate for screening but limits interpretation as a diagnostic model. Third, calibration error and subgroup heterogeneity suggest that further refinement may be needed before real-world implementation.

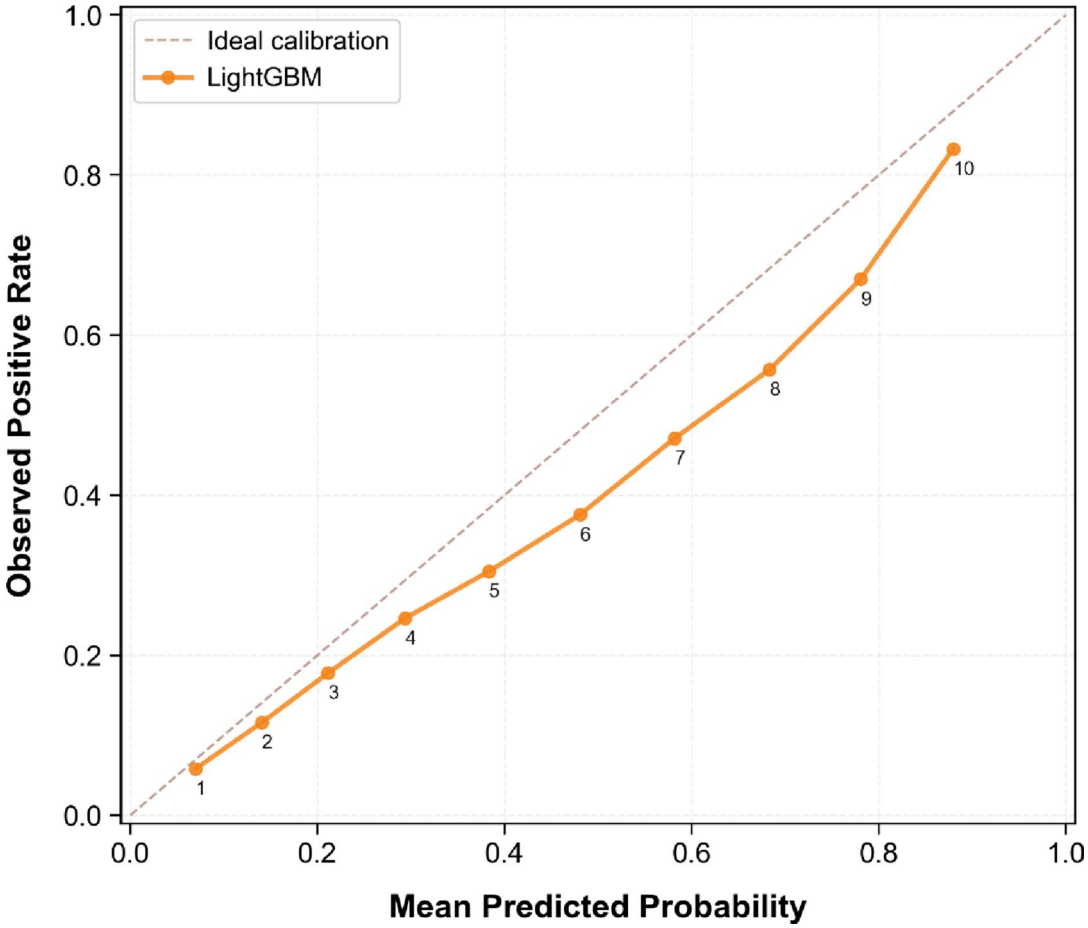


Fig. 4. Calibration LightGBM 2024 refined.

Subgroup Category	Subgroup	*N*	Positive Rate	AUROC	AUPRC	Accuracy	F1	Recall	Specificity	Brier Score
Sex	Male	17144	0.416	0.773	0.705	0.697	0.663	0.716	0.684	0.195
Sex	Female	9506	0.317	0.808	0.677	0.758	0.630	0.649	0.809	0.166
Age	< 40	15346	0.259	0.754	0.528	0.758	0.486	0.443	0.868	0.165
Age	40–59	10106	0.516	0.730	0.733	0.649	0.710	0.832	0.453	0.219
Age	≥ 60	1198	0.806	0.680	0.882	0.809	0.894	0.996	0.030	0.149
BMI	< 24	13559	0.280	0.786	0.613	0.764	0.565	0.547	0.848	0.162
BMI	24-27.9	9203	0.460	0.750	0.716	0.674	0.677	0.743	0.615	0.208
BMI	≥ 28	3201	0.579	0.737	0.787	0.661	0.753	0.889	0.348	0.213

Table 4. Subgroup Performance of the Main LightGBM Model in the 2024 Temporal Validation Cohort. Performance was evaluated in prespecified sex, age, and BMI subgroups within the independent 2024 temporal validation cohort. AUROC, area under the receiver operating characteristic curve; AUPRC, area under the precision-recall curve.

In conclusion, this study suggests that CBC-based machine learning, particularly LightGBM, may provide a practical approach for identifying individuals at elevated risk of diabetes in routine health examination settings. The model showed stable overall performance under temporal validation and may serve as an opportunistic screening aid to support further confirmatory testing. External validation in other institutions and prospective evaluation will be needed before broader clinical application.

Data availability

Due to intellectual property restrictions and potential future research applications, the code and data from this study are not publicly available. However, School of Biomedical Engineering, Shenzhen University Medical School is committed to advancing scientific knowledge and welcomes inquiries for collaborative research or to

gain a deeper understanding of the study. Access to the code and data may be provided under a data-sharing agreement. Please contact Guo Dan at [danguo@szu.edu.cn](mailto: danguo@szu.edu.cn) for further information or to initiate a discussion.

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References

1. (WHO). W. H. O. *Diabetes*, (2024). https://www.who.int/news-room/fact-sheets/detail/diabetes
2. Galicia-Garcia, U. et al. Pathophysiology of type 2 diabetes mellitus. *Int. J. Mol. Sci.* **21**, 6275 (2020).
3. Giri, B. et al. Chronic hyperglycemia mediated physiological alteration and metabolic distortion leads to organ dysfunction, infection, cancer progression and other pathophysiological consequences: An update on glucose toxicity. *Biomed. Pharmacother.* **107**, 306–328 (2018).
4. Duncan, B. B., Magliano, D. J. & Boyko, E. J. IDF diabetes atlas 11th edition 2025: global prevalence and projections for 2050. *Nephrol. Dial Transpl.* https://doi.org/10.1093/ndt/gfaf177 (2025).
5. Cavan, D. Why screen for type 2 diabetes? *Diabetes Res. Clin. Pract.* **121**, 215–217 (2016).
6. Magliano, D. J. & Boyko, E. J. IDF diabetes atlas. (2021).
7. Society, C. D. Guideline for the prevention and treatment of type 2 diabetes mellitus in China (2020 edition). *Chin. J. Diabetes Mellitus.* **13**, 315–409 (2021).
8. Sun, H. et al. IDF Diabetes Atlas: Global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res. Clin. Pract.* **183**, 109119 (2022).
9. Gutzeit, A. et al. Breast cancer in Switzerland: a comparison between organized-screening versus opportunistic-screening cantons. *ESMO open.* **9**, 103712 (2024).
10. Cheng, X. et al. Opportunistic screening using low-dose CT and the prevalence of osteoporosis in China: a nationwide, multicenter study. *J. Bone Miner. Res.* **36**, 427–435 (2020).
11. Jaiswal, V., Negi, A. & Pal, T. A review on current advances in machine learning based diabetes prediction. *Prim. Care Diabetes.* **15**, 435–443 (2021).
12. Ganie, S. M., Malik, M. B. & Arif, T. Performance analysis and prediction of type 2 diabetes mellitus based on lifestyle data using machine learning approaches. *J. Diabetes Metabolic Disorders.* **21**, 339–352 (2022).
13. Pei, D., Zhang, C., Quan, Y. & Guo, Q. Identification of potential type II diabetes in a Chinese population with a sensitive decision tree approach. *Journal of diabetes research* 4248218 (2019). (2019).
14. Seo, I. H. & Lee, Y. J. Usefulness of complete blood count (CBC) to assess cardiovascular and metabolic diseases in clinical settings: a comprehensive literature review. *Biomedicines* **10**, 2697 (2022).
15. Szablewski, L. & Sulima, A. The structural and functional changes of blood cells and molecular components in diabetes mellitus. *Biol. Chem.* **398**, 411–423 (2017).
16. Tahir, A. et al. Fingerprinting hyperglycemia using predictive modelling approach based on low-cost routine CBC and CRP diagnostics. *Sci. Rep.* **14**, 1090 (2024).
17. Mansoori, A. et al. Prediction of type 2 diabetes mellitus using hematological factors based on machine learning approaches: a cohort study analysis. *Sci. Rep.* **13**, 663 (2023).

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Author contributions

X.G. and G.D. contributed to Conceptualization and Project administration. X.G. was responsible for Methodology, Data curation, Software, and Writing – original draft. X.Y. contributed to Supervision. J.G. contributed to Methodology and Data curation. N.Z. and B.L. contributed to Resources. All authors (X.G., N.Z., B.L., X.Y., J.G., and G.D.) contributed to Formal analysis, Validation, Writing - Review & Editing, and Visualization, J.G. and G.D. contributed to Funding acquisition.

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Declarations

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethics statement

This study was conducted in accordance with the Declaration of Helsinki. The study protocol was reviewed and approved by the Ethics Committee of Shenzhen University, Shenzhen (Approval No. KYLL-20240403 A). As this was a retrospective study involving only fully anonymized and de-identified routine clinical data, the Ethics Committee waived the requirement for written informed consent.

Additional information

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